

Torqen: Cost-Aware Maintenance Decision Intelligence for Industrial Machinery

A Cost-Sensitive Decision-Support Framework Combining Failure Prediction, Diagnosis, and Maintenance Prioritization

Snigdha Routray

Manipal Institute of Technology

Abstract

Most of the predictive maintenance algorithms stop at the prediction stage of equipment failure without taking into account what should be done following such predictions. This report presents Torqen, a cost-aware maintenance decision support system that includes failure prediction, failure diagnostics, decision making based on expected costs (MONITOR / SCHEDULE / STOP), prioritization with consideration of maintenance capabilities, and simulation-based comparison of decision-making policies. The experiments carried out on the artificial predictive maintenance data set (AI4I 2020, with 10,000 samples and 3.39% failures) show the various performance (AUC=0.897 – 0.988) of three classification algorithms: Logistic Regression, Decision Tree, and Random Forest and prove that the impact of decision-making is determined not only by classification performance but also by the stability of decisions made based on the predicted probabilities. The classifier with high-AUC that makes predictions using discrete probabilities (Decision Tree) results in extremely volatile decisions in case of use for decision making based on expected cost computation, while the one that predicts using smoothed probabilities (Random Forest) can help construct an applicable decision strategy. In addition, we get the cost-optimal decision threshold $t^*=0.04$ that allows saving almost 4 times more costs than in case of use of the usual 0.5 threshold and show that Torqen saves 77.0% of cost compared to reactive maintenance and 73.4% of cost in comparison with scheduled maintenance in a 52-week simulation for fleet. The larger is the assumed downtime cost, the better the performance.

Keywords—*predictive maintenance, cost-sensitive classification, decision support, Industry 4.0, maintenance scheduling, RapidMiner*

I. INTRODUCTION

Unexpected downtime continues to be one of the main sources of costs in discrete manufacturing operations. Studies conducted by the industry estimate the cost of unexpected downtime in the order of tens to hundreds of thousands of monetary units per hour on the facility level, while academic studies indicate that the estimated global losses from unexpected downtime in manufacturing operations amount to tens of billions of dollars per year [1], [2]. Predictive Maintenance (PdM) has become a prevalent solution: being able to predict when a machine will break down, the company will move from repair to maintenance.

Nevertheless, current literature and tools on PdM largely remain at the level of classification - determining whether the machine is going to fail without solving the problem of how the maintenance manager is supposed to use this information, what the cost implications are, and within which constraints. The model that possesses high ability to rank (high AUC value) will yield a suboptimal operational strategy, if its probabilistic predictions are not precise enough for a good decision making and the threshold is set up arbitrarily (0.5).

This report attempts to close this gap by proposing Torqen, a decision-intelligence module that sits atop PdM classifiers and does the following:

1. forecasts machine breakdowns
2. identifies the cause of possible malfunction
3. uses an expected-cost optimization strategy to translate predictions into maintenance actions (either MONITOR, SCHEDULE, or STOP)
4. performs prioritization subject to realistic constraints on maintenance capacity
5. is judged not only on its performance in prediction tasks but also on simulated costs in comparison with traditional maintenance strategies.

II. PROBLEM STATEMENT

Could there be predictive or diagnostic machine-learning models for the detection of breakdowns in industrial machines that, based on failure costs, maintenance costs, decision thresholds, and the limited capability for maintenance, suggest actions for minimizing the operational cost?

III. OBJECTIVES

1. Predict failure based on operational data and sensors.
2. Identify the possible cause of failure based on the predicted failure or actual failure.
3. Design an algorithm that determines the maintenance to be taken based on risk prediction, where costs are considered in the process.

4. Prioritize the machines to maintain using maintenance capacity constraint optimization.
5. Simulate Torqen compared to Reactive and Fixed-Schedule maintenance strategies.

IV. RELATED WORK

Studies on predictive maintenance include research on failure prognostics, remaining useful life estimation, and fault diagnosis, and are usually analyzed through traditional classification performance measures, including accuracy, precision, recall, and AUC [3]. Cost-sensitive classification involves assigning different penalties based on the consequences of the misclassification instead of considering all misclassifications equal [4]. On the other hand, the maintenance optimization literature considers constrained maintenance scheduling as a problem in scheduling or knapsack problem [5]. Decision support systems for maintenance are meant to link prediction and decision-making; however, only a handful of systems in literature that involve failure prediction, multi-mode fault diagnosis, an explicit expected cost criterion for decision-making, and priority assignment with capacity constraints have been reported in the literature.

V. DATASET

The proposed work employs AI4I 2020 Predictive Maintenance Dataset that is a synthetic but realistic benchmark capturing the characteristics of real predictive maintenance data [6], [7]. This dataset consists of 10,000 observations on machines with 14 features without any missing value. Machine failures are highly imbalanced where there are 339 positive (failed) observations out of 10,000 (3.39%). There are five binary indicators for the different failure modes of the machine: Tool Wear Failure (TWF, $n = 46$), Heat Dissipation Failure (HDF, $n = 115$), Power Failure (PWF, $n = 95$), Overstrain Failure (OSF, $n = 98$), and Random Failure (RNF, $n = 19$).

Since this dataset is a synthetic benchmark, all findings should be interpreted only in terms of the proof of concept and comparative analysis.

A. Failure-Label Structure

It is noted that the total number of positive counts of each failure mode ($46+115+95+98+19=373$) is greater than the total number of positive counts of Machine-failure (339). It shows that there is no mutual exclusivity of the failure modes: some machines have multiple flags of failure mode at the same time. In other words, the failure mode identification is a multi-label problem (Section IX).

VI. METHODOLOGY

A. Data Leakage Controls

Both UDI and Product ID will be removed from consideration as predictor variables. The key point here is that the five failure mode predictors (TWF, HDF, PWF, OSF, and RNF) will not be included as predictors in predicting Machine Failure.

B. Feature Engineering

There are two derived attributes that are created through domain reasoning, namely:

- 1) temperature difference = Process temperature – Air temperature
- 2) torque-speed product = Torque $\times$ Rotational speed

which is described as a power attribute proxy and not a power attribute after conversion into units. The above attributes are created natively in Altair AI Studio using Generate Attributes operator.

C. Model Training and Validation

Three types of classifiers namely logistic regression, decision tree, and random forest are built and analyzed using the 10-fold cross validation technique within Altair AI Studio. The class imbalance is tackled mainly through cost-based thresholding (Section VIII) and not through resampling techniques.

D. Cost Model

Three types of maintenance activities are stated as follows: MONITOR (do nothing), SCHEDULE (maintenance), and STOP (shutdown). With respect to the estimated failure probability $P(\text{failure}|x)$, the cost of action a is $EC(a|x) = P(\text{no failure}|x) \cdot C(a, \text{no failure}) + P(\text{failure}|x) \cdot C(a, \text{failure})$, and the optimal action a^* is $a^* = \text{argmin}_a EC(a|x)$. The cost coefficients (see Table I) are assumed explicitly and are loosely based on industry downtime-cost standards [1], [2] normalized to individual machine costs.

TABLE I. ASSUMED COST MATRIX (ARBITRARY CURRENCY UNITS)

Action	Actual: No Failure	Actual: Failure
MONITOR	0	18,000
SCHEDULE	300	800
STOP	600	400

E. Maintenance Capacity Constraint

Machines that can be recommended for SCHEDULE and STOP are ordered based on their expected savings (savings from using these strategies compared to MONITOR) and a maximum number K is served within each maintenance period.

F. Policy Simulation

In a 52-week, 200-machine simulation experiment, there are comparisons among the following three policies: Reactive policy (without monitoring, complete emergency cost of repair per machine failure), Fixed Schedule (all machines being serviced every 4 weeks irrespective of their risk level), and Torqen (risk-ranking, threshold and capacity-limited interventions). Resampling is done during each period from the actual data (probability-prediction, outcome-pair) of the trained model for each machine, with replacement, in order to retain the true correlation between predicted risk and outcome and expand the cross-section AI4I dataset into a simulation dataset over time periods (Section XIII).

G. Tooling

Importing of data, preprocessing, feature engineering, modeling, validation and extracting of feature importance are implemented in Altair AI Studio (RapidMiner). The decision rule based on expected costs, threshold optimization, failure mode analysis, prioritization with respect to constraints and policy simulation are performed in Python. Also, the core decision rule MONITOR/SCHEDULE/STOP is implemented natively in Altair AI Studio through the application of Generate Attributes operator of the form `recommended_action = if(confidence(1) < 0.04, "MONITOR", if(confidence(1) < 0.15, "SCHEDULE", "STOP"))` directly to the probability output by the Random Forest model, using the cost-optimal threshold $t^* = 0.04$ computed in section X-B.

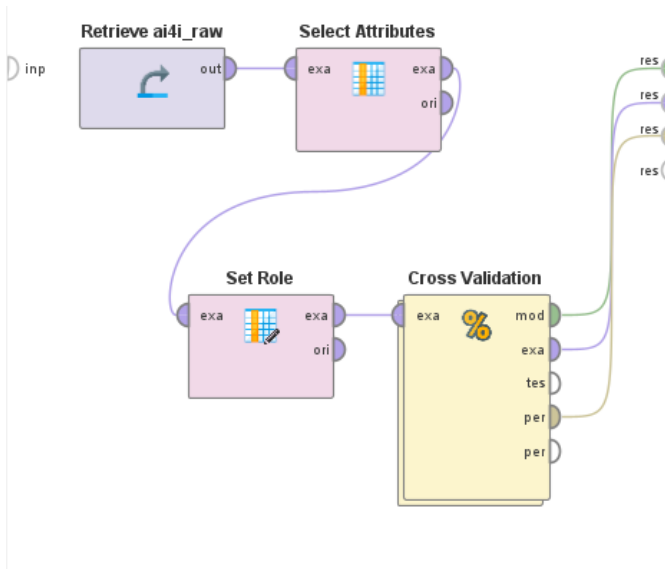


Fig. 1. Torqen RapidMiner process overview (data import through model training).

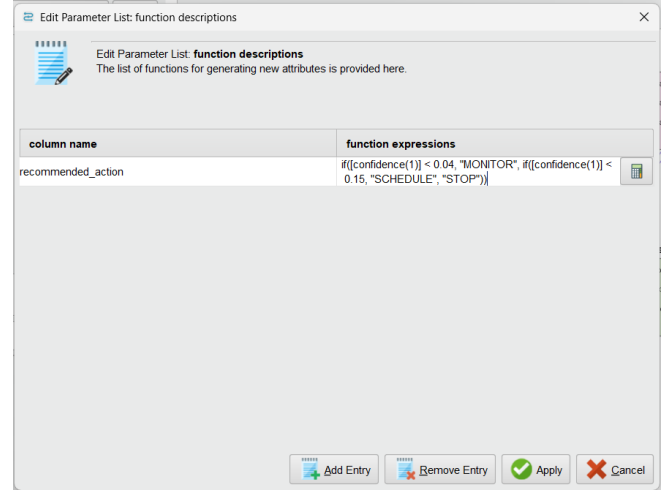


Fig. 2. Native RapidMiner implementation of the cost-based MONITOR/SCHEDULE/STOP decision rule.

VII. EXPLORATORY DATA ANALYSIS

Dataset verification confirmed 10,000 rows, 14 attributes, and zero missing values across all columns. Numeric feature ranges are physically plausible: air temperature 295.3–304.5 K, process temperature 305.7–313.8 K, torque 3.8–76.6 Nm, tool wear 0–253 min, rotational speed 1168–2886 rpm.

Machine Failure Class Distribution (AI4I 2020, n=10,000)

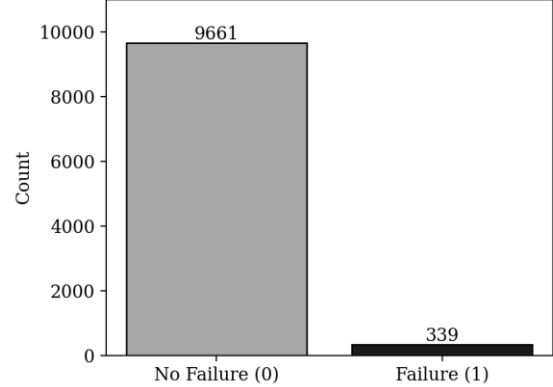


Fig. 3. Machine failure class distribution, illustrating the 3.39% positive-class imbalance.

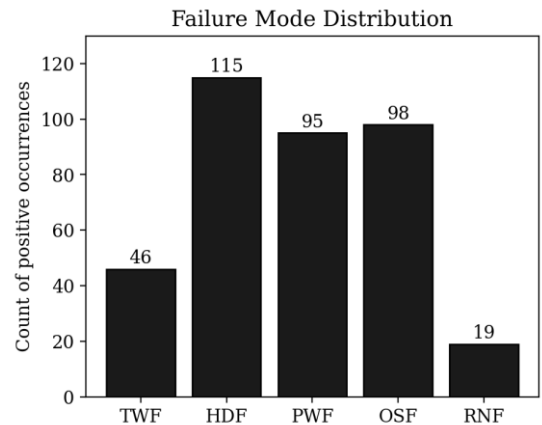


Fig. 4. Failure-mode distribution across the dataset.

VIII. MODEL COMPARISON RESULTS

Table II presents 10-fold cross-validated results for the three baselines. Decision Tree and Random Forest yield almost perfect AUC (ranking performance), but relatively low recall for the class “failure” at the standard threshold of 0.5 – clear indication that accuracy and even AUC are not enough to describe the utility of a model in a case of class imbalance (Section X).

TABLE II. MODEL COMPARISON (10-FOLD CROSS VALIDATION)

Model	Acc.	AUC	Prec.(1)	Rec.(1)	F1(1)
Logistic Regression	96.38%	0.897	46.81%	49.85%	48.46%
Decision Tree	97.33%	0.988	86.73%	25.07%	38.88%
Random Forest	97.25%	0.950	84.04%	23.30%	36.46%

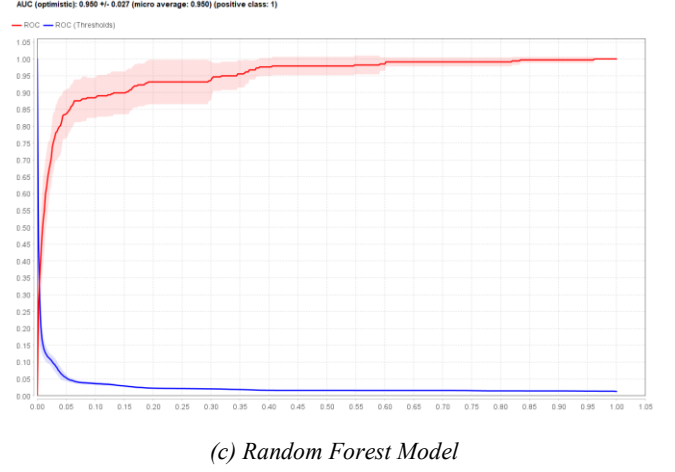
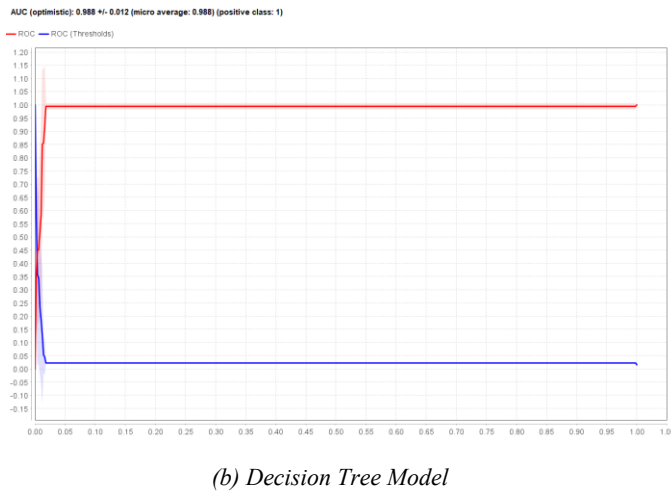
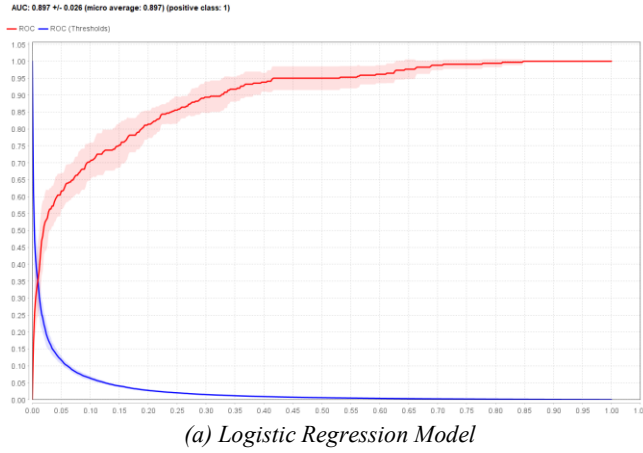


Fig. 5. ROC curves for the three baseline classifiers: (a) Logistic Regression (AUC=0.897), (b) Decision Tree (AUC=0.988), (c) Random Forest (AUC=0.950).

A. Feature Engineering Impact

The inclusion of temperature_difference and torque_speed_product resulted in an increase in accuracy from 97.25% to 97.73% and most importantly, in the recall score on failure from 23.30% to 33.92%. In terms of feature importance according to the Gini-Index (see Table III), the torque_speed_product is ranked higher than the raw Torque, while the temperature_difference is ranked higher than both raw temperatures.

TABLE III. FEATURE IMPORTANCE (GINI INDEX)

Rank	Feature	Weight
1	torque_speed_product	0.012
2	Torque [Nm]	0.008
3	Rotational speed [rpm]	0.005
4	temperature_difference	0.003
5	Tool wear [min]	0.003
6	Air temperature [K]	0.001
7	Process temperature [K]	0.000
8	Type	0.000

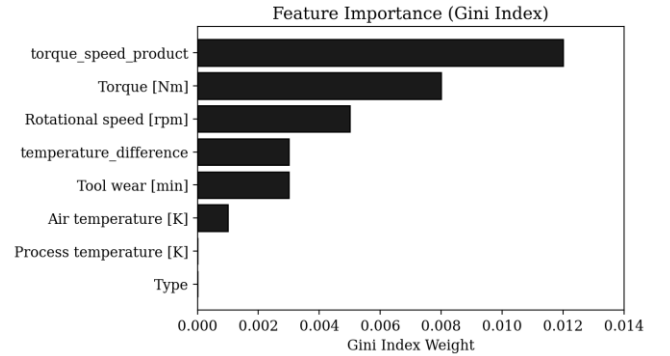


Fig. 6. Feature importance ranking, Random Forest with engineered features.

IX. FAILURE-MODE DIAGNOSIS

Considering the similarities between labels identified in Section V-A, the diagnosis process involves training five independent class balanced Random Forest classifiers for each failure type based solely on the 339 instances for which Machine failure = 1, using a 70-30 train-test split ratio.

TABLE IV. FAILURE-MODE DIAGNOSIS RESULTS (n=339 FAILURES)

Mode	Support	Prec.	Rec.	F1
TWF	46	0.917	0.786	0.846
HDF	115	0.971	0.971	0.971
PWF	95	1.000	1.000	1.000
OSF	98	0.967	1.000	0.983
RNF	1	N/A	N/A	N/A

The accuracy of diagnosis increases according to class support: PWF and OSF classes have diagnoses that are nearly perfect, while HDF is diagnosed quite well, and TWF less so (n=46). RNF is represented by just one instance in reality out of 19 cases found in a full 10,000-line dataset – thus proving that RNF is basically unrelated to the Machine’s failure condition, which is how the documentation of AI4I explains the inclusion of RNF, a random sensor-agnostic class introduced to detect overfitting to noise.

X. COST-AWARE DECISION MAKING

A. Probability Granularity Finding

The first cost-model run took probabilities from the Decision Tree, which tend to form a small set of discrete leaf-node probabilities (e.g., 0.977/0.023 repeatedly). With 5,000 cost for missing failure, it yielded near-complete under-intervention (83 out of 102 failed machines were missed, virtually all machines defaulted to MONITOR); with the cost increased to 18,000 and the same Decision Tree probabilities, it yielded complete over-intervention (2,970 out of 3,000 machines were recommended for intervention, only 6 were missed, but with an impractical 99% intervention rate). Taking Random Forest probabilities instead – smoother, since averaged over multiple trees – fixed this instability, producing a gradation and practical decision distribution (2,322 MONITOR / 651 SCHEDULE / 27 STOP) under the same cost assumptions, with 94.1% of failures found at the cost of 20.1% false positives. This is mentioned as one of the key methodological results: not only classifier impacts ranking quality (AUC), but its stability and usability in cost-based decisions.

B. Threshold Optimization

Raising the decision threshold from 0.01 to 0.50 in the probabilities provided by Random Forest (see Fig. 7) produces an optimal threshold of $t^* = 0.04$, which results in 87.2% recall with the total cost being 401,700 compared to 17.6% recall with the total cost of 1,525,500 for the standard $t = 0.50$ threshold, despite $t = 0.50$ threshold having higher

precision. This proves that the thresholds chosen to optimize the standard classification measures may be far from optimal for the business problem.

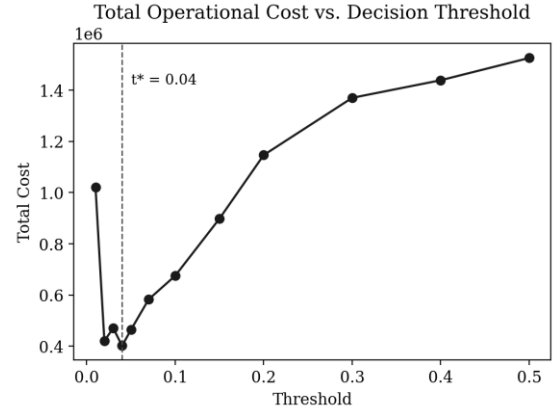


Fig. 7. Total operational cost as a function of decision threshold; minimum at $t^*=0.04$.

XI. MAINTENANCE CAPACITY PRIORITIZATION

Out of 3,000 test machines, 678 machines fall under the category of SCHEDULE and/or STOP. When these are ranked based on their expected savings and the top K machines are maintained in each time window (Table V), it clearly reflects the expected behavior of diminishing returns, where marginal saving is decreasing with increasing capacity.

TABLE V. CAPACITY SENSITIVITY (K = MACHINES SERVICEABLE PER WINDOW)

K	Saving Captured	% of Ceiling
20	319,618	34.2%
50	490,717	52.5%
100	624,509	66.8%
200	782,679	83.7%
678 (ceiling)	935,005	100.0%

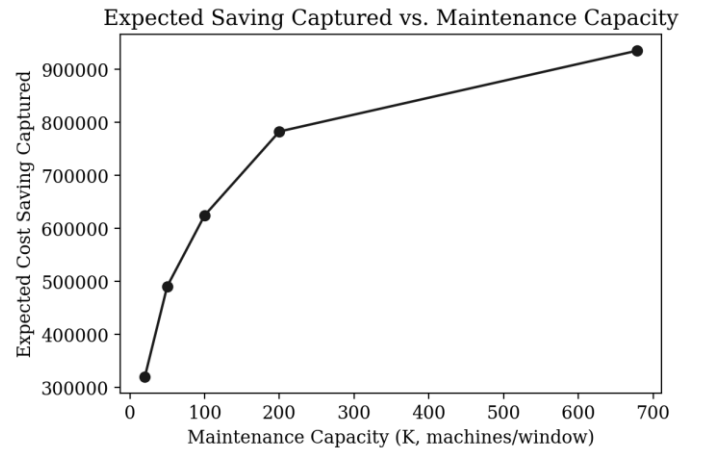


Fig. 8. Diminishing returns of maintenance capacity on total expected saving.

XII. MAINTENANCE POLICY SIMULATION

A 52-week, 200-machine simulation fleet (VI-F) of the Reactive, Fixed Schedule, and Torqen strategies is conducted (VI table). The total cost incurred for the Torqen strategy is 77.0% lower than that for the Reactive strategy and 73.4% lower than that for the Fixed Schedule strategy, considering 19.5 machines on average being served every period (limited to 20) and 84.5% of total failures identified.

TABLE VI. 52-WEEK POLICY SIMULATION RESULTS

Policy	Total Cost	Uncaught Fail.
Reactive	6,714,000	373
Fixed Schedule	5,820,000	280
Torqen	1,547,100	58

**Total Simulated Maintenance Cost by Policy
(52 Weeks, 200 Machines)**

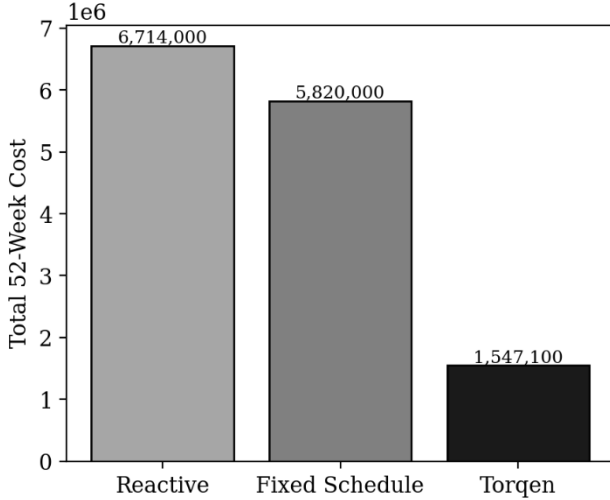


Fig. 9. Total 52-week simulated cost by maintenance policy.

A. Sensitivity Analysis

Repeating Table VII under the Low, Base, and High costs for emergency failure conditions, we can see that Torqen's cost advantage persists in all three cases and grows as the assumed cost of downtime grows (69.5% $\rightarrow$ 77.0% $\rightarrow$ 80.0% savings vs. Reactive). This is a favourable robustness result since high costs of downtime are a conservative assumption for critical machinery.

TABLE VII. SENSITIVITY ANALYSIS: TORQEN SAVINGS VS. REACTIVE

Scenario	Emerg. Cost	Torqen Savings
Low	9,000	69.5%
Baseline	18,000	77.0%
High	30,000	80.0%

**Robustness of Torqen Cost Advantage
Across Downtime-Cost Scenarios**

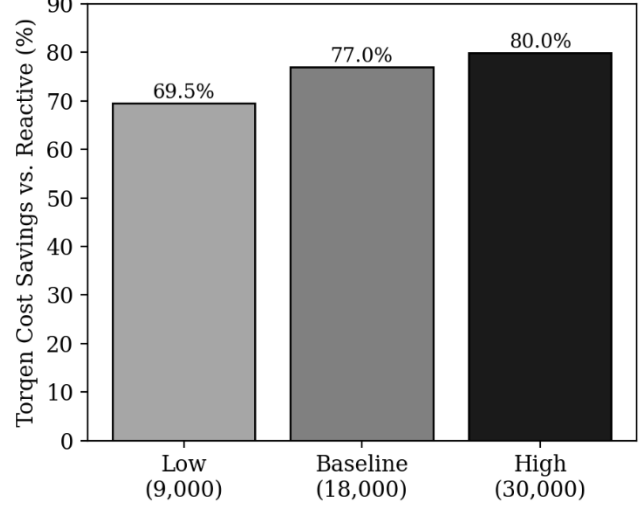


Fig. 10. Robustness of Torqen's cost advantage across downtime-cost scenarios.

XIII. DISCUSSION

The following are the three most remarkable conclusions from the presented analysis. First, AUC alone cannot be used as a metric for assessing the suitability of the particular model for cost-based decisions since, despite the better AUC value of the Decision Tree (0.988 compared to 0.950 for the Random Forest), the former was unable to provide an applicable decision policy owing to the large variances of probability estimations resulting in the unstable binary decisions of the model according to the expected cost criterion. Secondly, the cost-optimal decision threshold (0.04) is significantly smaller not only than the default 0.5 but also than the threshold obtained in the F1 score maximizing process; obviously, optimizing the F1 score and reducing costs is two separate problems which may be drastically different in cases when the classes are imbalanced. Lastly, the simulated cost savings of Torqen over Reactive (77%) are significant, although partly due to the particular cost matrix used.

XIV. LIMITATIONS

1. The AI4I is an artificial benchmark, whereby the results are based on machine operation/sensor readings and model performance and not real telemetry and outcomes from a validated real factory.
2. The policy simulation (Section VI-F, XII) makes use of cross-sectional resampling to generate a longitudinal fleet, because the AI4I dataset has no longitudinal data for each machine; this preserves the realistic (probability, outcome) pair but not the actual dynamic process such as the degradation process and maintenance process that causes a change in state.

3. It was not possible to model RNF diagnosis due to the single positive example of actual failure cases.
4. Only sensitivity of emergency failure cost parameter is considered, whereas capacity and threshold parameters are held constant.

XV. PRACTICAL IMPACT

The action plan generated by Torqen is not simply a risk score but a prioritized and capacity-sensitive plan of actions for a maintenance manager where each machine is classified as either MONITOR, SCHEDULE, or STOP, accompanied with an explanation (such as expected mode of failure along with the readings from related sensors) and sorted by decreasing value such that a single maintenance team can be used to take care of the most valuable ones first. The capacity sensitivity output (Table V) further provides management with a clear decision-making tool regarding capacity planning.

XVI. FUTURE WORK

Priority extension problems involve:

1. Validation against actual data from factories, including maintenance records.
2. Joint sensitivity analysis on cost, threshold, and capacity.
3. Formal capacity allocation optimization (using the knapsack approach), replacing the current priority queue.
4. Real time/stream processing.
5. Cross-validation against another predictive maintenance benchmark dataset to verify generalizability of model results as well as decision layer results.

XVII. CONCLUSION

The contributions of this report include Torqen, which is a cost-aware maintenance decision support framework that integrates failure prediction with diagnosis, cost decision making, prioritization under capacity constraint, and simulation of policy assessment. In addition to the main finding of 77% cost reduction compared with reactive maintenance and being resilient to different cost assumption scenarios in simulation, the core methodological contribution of this report is the fact that model's stability for cost-based decisions relies on probability calibration and granularity, and not on ranking only.

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